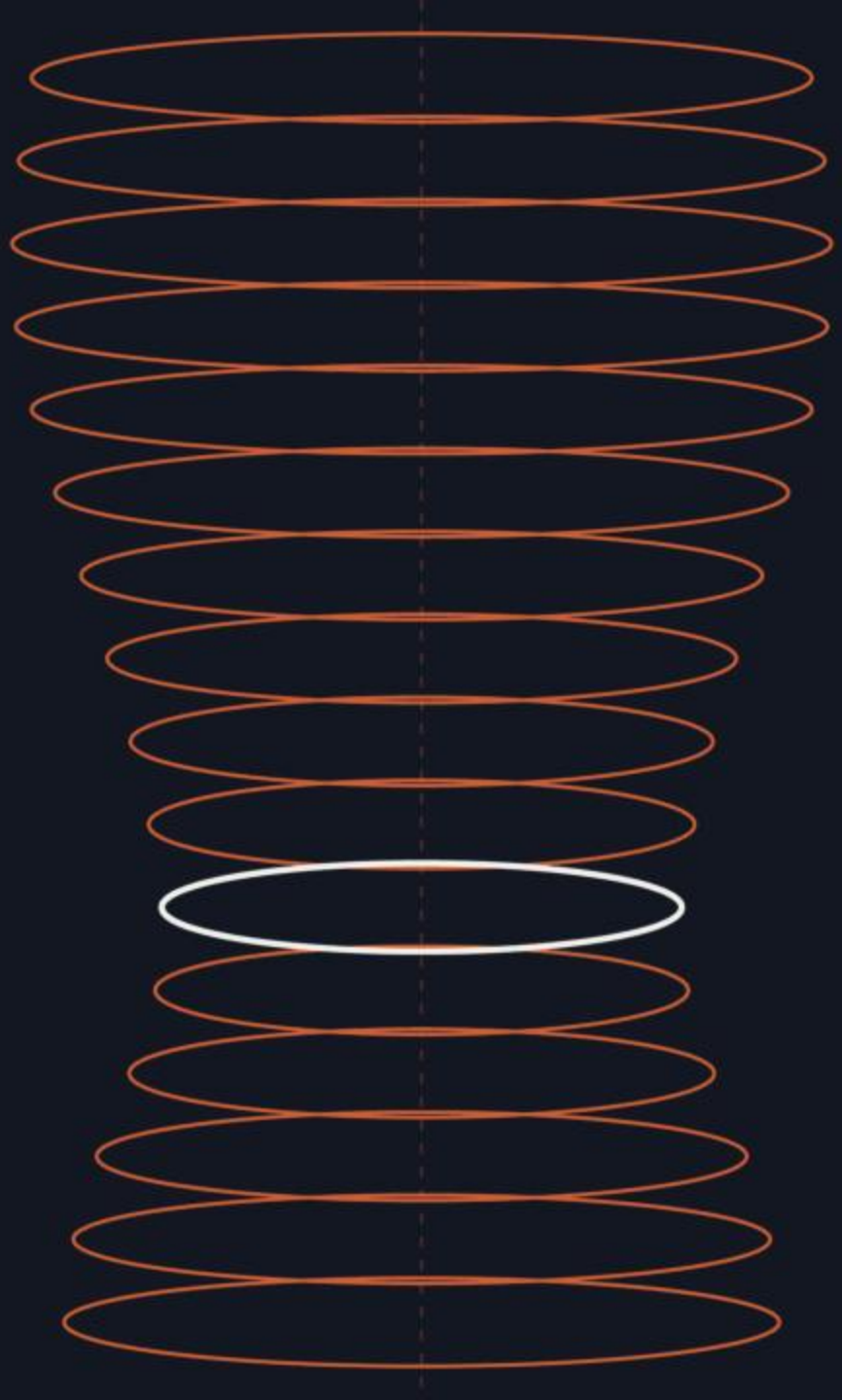
Fitted

Photo-based measurement and CAD customization for prosthetics

Nicole Ma, Danica Sun, Shriya Sreeju



Volunteers fill a gap the health system leaves

e-NABLE, an international 501(c)3 connects people with upper-limb differences to volunteers who create free, customized, 3D-printed assistive hands and arms.

Many recipients are children.

The primary constraint? Volunteers spend

6–20 hours

designing each prosthetic sized to measurements

[1] WHO/UNICEF Global Report on Assistive Technology, 2022 · [2] enablingthefuture.org ·

Women already get less from prosthetic care

33%

longer wait than men for a first prosthesis[1]

43 vs 69%

of women vs men fitted with a prosthesis by discharge[2]

<5%

of participants in veterans' prosthetic studies are women[3]

For upper-limb devices, we found **no data split by sex at all.**

Lower-limb clinical studies: [1] U. Michigan, J Clin Transl Sci 2024 abstract (claims 2015–21) · [2] Singh et al., Disabil Rehabil 2008, n=105 · [3] Kuo et al., PM&R 2023

One number can’t fit her hand’s shape

Women’s hands are **7% shorter** but **11% narrower** than men’s, but eNABLE base designs are uniformly scaled.



Her hand

78 × 180 mm

Stretched to her width

≈9 mm short

Stretched to her length

≈4 mm too wide

Fitted turns phone photos or LiDAR scans into a customizable e-NABLE hand sized in width and length.

The process

CLAUDE
judgement calls

Photo check
Rejects blur and hidden markers; explains the retake

Landmarks
Finds the wrist crease and gauntlet ends on a mm grid

Choose device
e-NABLE rules; "stars" becomes a filament spec

↻ ≤ 3 fix-up passes

Render review
Critiques renders; may nudge only 4 settings

CODE

Input
3 phone photos or a LiDAR scan

Straighten
ArUco markers → homography → true mm; SAM 2 cut-out

Measure, mesh
Width + thickness every 5 mm → ellipse slices → closed mesh

Generate
OpenSCAD hand, width ≠ length, joints intact; distance-field gauntlet

Printability
Watertight + wall checks; PrusaSlicer; colored 3MF

Volunteer
Signs off; 15-min fit-test ring, then the full print

CUSTOM



Width × length
scaled per hand



Part colors
from real filament



Raised pattern
Stars, hearts, etc

Accurate across sex, age and skin tone

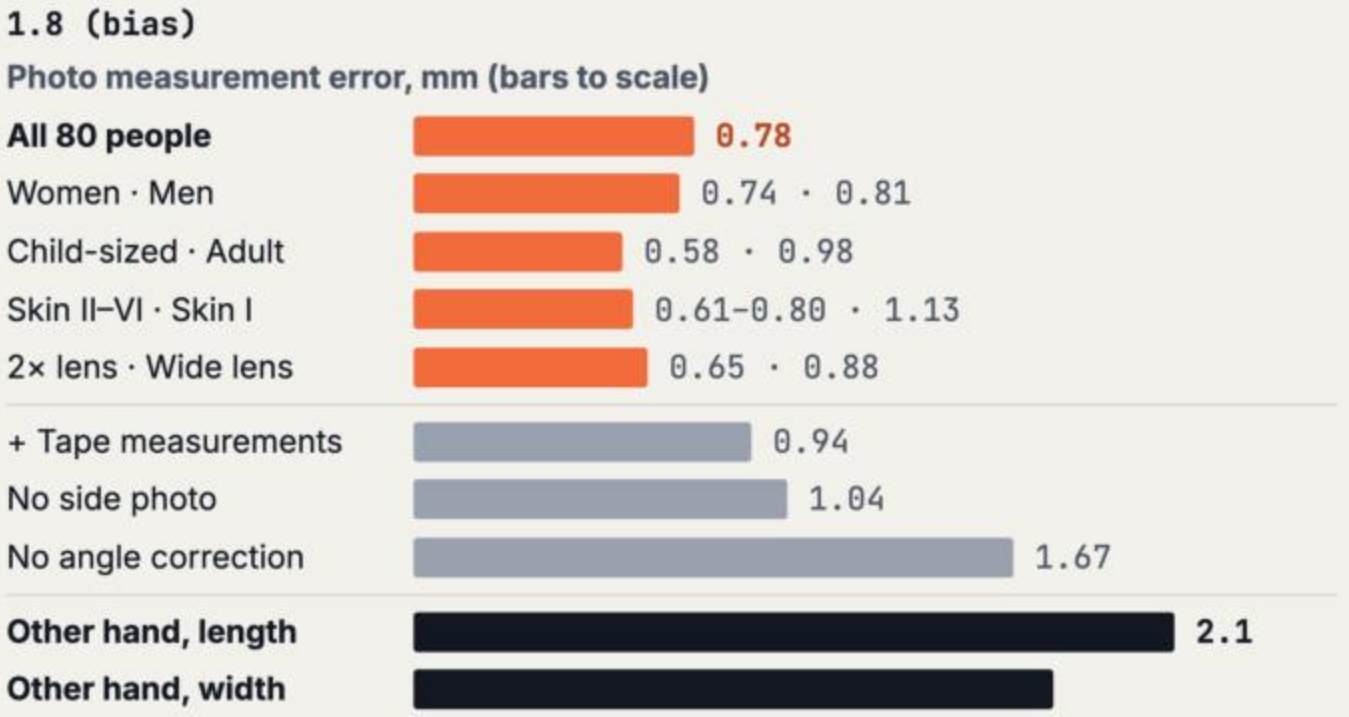
PHOTOS · 80 PEOPLE
40 women, 40 men from ANSUR II; half child-sized; skin I–VI; 55–110 cm, wide and 2× lens

STEPS TURNED OFF
Same photos without the side view or angle correction, or with tape added

LIDAR · 30 SCANS
Noise and holes, random poses: 0.47 mm, 100% within 2 mm

SIZING · 6,068 HANDS
Every ANSUR II hand; width within 3 mm for 100% of women and men

PHOTO CHECK · 24 PHOTOS
All 20 bad ones caught; no false alarms on 4 good



"Child-sized" = adult ANSUR II hands shrunk to 70–88%, not real children. Two-value rows: bar shows the first.

Under a millimetre off, and women’s length error cut 4×

80

synthetic people built from real U.S. Army measurements

0.78 mm

average error from photos, with 91% within 2 mm

10.4 → 2.4

mm of women’s hand-length error, sizing width and length separately

*half women, half child-sized, all six skin tones

“2–3 mm is the difference between a good and bad fit”[1] -eNABLE

Inside the 2–3 mm fit window, for women too



Bars to scale, 60 px per mm. HandSizer: e-NABLE Hub wiki. HandAnthro: arXiv 2609.37855, 2026 preprint.

FITTED

Sized for her.

Every 100 devices take **600–2,000 volunteer hours** to design today. Fitted is built to give those hours back, and to fit her the first time.

Limitations



<5%

The data leaves out women and children

Women: under 5% of veterans' prosthetic study participants, 4 of 33 OpenLimbTT scans.



4–10 mm

The designs assume one hand shape

Even scaling misses by 4–10 mm against a 2–3 mm fit window. We scale width and length separately; stock joints still need strength testing.



0.3 mm

Photo accuracy depends on the photo

0.3 mm on the forearm in rendered photos of one synthetic arm; Real photos add lighting, angle and phone error.



Fit isn't just size

Comfort, padding, cord tension and grip still need a volunteer and a fitting session. A volunteer approves every file before printing.

Sources: Kuo et al., PM&R 2023 (lower limb, veterans) · OpenLimbTT (lower-limb scans) · ANSUR II (U.S. Army adults)

Next Steps

NOW



Pilot with a real chapter

Size devices for real recipients with the e-NABLE chapter at teammate Nicole Ma's former high school.

3 MONTHS



Measure the gender gap

An opt-in dashboard of request-to-delivery time by gender. No one tracks this yet.

6 MONTHS



Build the missing dataset

With consent, turn pilot measurements into an open hand dataset of women and children.

6 MONTHS



Grow with the kid

Re-generate the device from new photos as a child grows, instead of re-sizing by hand.

1 YEAR



Proportion-aware designs

With e-NABLE designers: Phoenix v3, Kinetic and Osprey that scale width and length separately, joint-tested.

Appendix A · Prior art

Tool	Input	Output	Limit
Blender method, sizing spreadsheet	Hand measurements	One scale %	Spreadsheet “may not produce a properly fitting device”
Handomatic	Hand measurements	Pre-scaled STLs	One scale (team research)
e-NABLE AI HandSizer	Sound-hand photo with a card	Scale %, Phoenix v3 only	~12 mm average error, designer’s own tests
AI parametric hand generator	Typed measurements	LLM-suggested settings	No photos or scans; no gauntlet fit
HandAnthro (preprint)	One palm-up phone photo	44 hand dimensions	3.80 mm mean error; no device
Ours	Photos or LiDAR on a marker sheet	Width + length scales, fitted gauntlet, fit ring	Accuracy not yet measured

Appendix B · How we'll evaluate

ANSUR II check (synthetic)

1. Take women's 5th, 50th and 95th percentile hands, plus ~200 random women
2. Generate each hand two ways: uniform scale vs width + length scales
3. Compare to her ANSUR values: mean error, % within 2 mm
4. Repeat for men; chart uniform vs ours, by sex

Participant study

1. At least half women; caliper-measure knuckle width, palm length, wrist
2. Good photos plus deliberately bad ones
3. Report mean error by dimension and sex; bad-photo catch rate
4. Print uniform vs ours for one woman; film the fit

Limits: ANSUR II is U.S. Army adults, not children or recipients. We found no public 3D scans of partial-hand limbs. Teammates' intact forearms stand in for the gauntlet fit.

Appendix C · Judge questions

Your access stats are lower-limb?

Yes. They show a pattern; upper-limb data split by sex doesn't exist yet.

Are e-NABLE hands modelled on men?

Not documented. Any single base ratio misfits hands with other ratios.

Is it clinically validated?

No. It's a volunteer design tool; every draft needs volunteer approval.

Are you partnered with e-NABLE?

No. We build on its open designs and public sizing guidance.

ANSUR II is soldiers, not kids?

Right. It's a range check only; her own measurements drive the design.

What happens to the photos?

Processed locally, deleted after use. Claude sees renders and numbers only.

Appendix D · Sources

- WHO/UNICEF Global Report on Assistive Technology (2022): ~1 billion denied access
- enablingthefuture.org: ~40,000 volunteers, 100+ countries, est. 10,000–15,000 recipients
- Engineering for Change: e-NABLE device, $35 suggested price
- e-NABLE video: 6–20 h per device; 2–3 mm changes fit [link]
- e-NABLE Device Sizing: one scale %; spreadsheet “may not produce a properly fitting device”
- e-NABLE Hub sizing wiki: AI HandSizer, Phoenix v3, ~12 mm (designer’s tests)
- Paraglider / flexible_flyer, M. Mendenhall (GitHub)
- U. Michigan, J Clin Transl Sci 2024 abstract: women’s time ratio 1.33 (lower limb)
- Singh et al., Disabil Rehabil 2008: fitted at discharge 42.9% vs 68.6%, n=105
- Kuo et al., PM&R 2023: women <5% of veterans’ prosthetic study participants
- ANSUR II, U.S. Army 2012 (openlab.psu.edu/ansur2): medians via secondary summary
- Zhou et al., HandAnthro, arXiv 2609.37855 (2026 preprint): 3.80 mm
- Biddiss & Chau, Prosthet Orthot Int 2007: upper-limb rejection rates
- ai-parametric-prosthetic-hand-generator, P. Candeias (GitHub)